Mad Scientist 1

Four apparatus investigations for COP greater than one

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Abstract

Four open problems pursue an ambitious goal: a repeatable machine that delivers more useful energy than all the energy supplied to it. A falling-weight lever, switched transformer, contact chain and coupled-coil release supply four routes to investigate complete-cycle COP. Each experiment compares useful receiver work with all gross inward energy through preparation, operation and actual reset. Background, explanatory figures and exact illustrative mathematics connect each motivating observation to that test. The target is a reproducible COP lower bound above one with independently checked machine states.

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Local draft; the GitHub destination below is planned, not published.
Latest on GitHub: https://github.com/hobnilre/physics-mad-scientist-1

The top four

I want an apparatus that raises a load or charges a receiver, does it again, and leaves an energy record another experimenter can check. The ambition is a coefficient of performance (COP) greater than one: useful receiver energy divided by **all gross inward energy** over the complete history. Give the exciting output a receiver, give every input a meter, and see what survives.

This list alternates the two leading mechanical and two leading electrical investigations, preserving their order within each domain. The first pair starts the work; the second reuses much of its instrumentation. Rank sets experimental priority. The COP of each apparatus is a quantity to measure.

1. **The falling-weight lever.** Can timed compliant launch deliver more receiver work than the full lifting, release and reset input? [OP-CT-HW-01]
2. **The switched transformer.** Can physical tap commutation achieve COP above one with source, driver and reset inputs measured? [OP-TRF-02]
3. **The contact chain.** Can mass order, gaps and branching produce loaded COP above one over preparation, delivery and reset? [OP-LR54-03]
4. **The coupled-coil release.** Can timed release deliver more receiver work than complete energization, switching and reset require? [OP-LR48-03]

The starting points are striking: a fast launch, different voltages across different terminal pairs, and growing sums of momentum or flux-linkage magnitudes. Each suggests a physical idea worth following. Their open questions concern the contact histories and changing connections that turn the idea into an apparatus.

1 The falling-weight lever

A teeterboard turns the downward arrival of one or two people into the upward flight of another. A brief contact produces a dramatic launch, and changes in timing, leg compression or board flexibility can alter it. The falling-weight lever brings this contact-transfer idea to a tabletop, replacing performers with guided masses. Its COP question is whether timed contact can deliver more useful receiver work than preparing and restoring the apparatus requires.

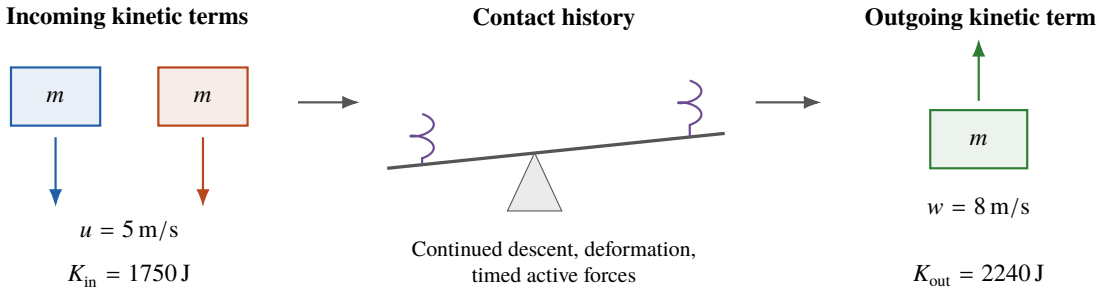


Figure 1. Illustrative endpoint comparison for three equal masses. The central mechanism is schematic; descent and deformation continue during contact.

The observation. In Figure 1, each mass is $m = 70\,\mathrm{kg}$. Two arrive downward at $u = 5\,\mathrm{m/s}$; one receiver leaves upward at $w = 8\,\mathrm{m/s}$. The selected kinetic terms give

$$K_{\mathrm{in}} = mu^2 = 1750\,\mathrm{J}, \quad K_{\mathrm{out}} = \frac{1}{2}mw^2 = 2240\,\mathrm{J}, \quad K_{\mathrm{out}} - K_{\mathrm{in}} = 490\,\mathrm{J}. \tag{1}$$

The selected kinetic ratio is $2240/1750 = 32/25$. Turning that motivating comparison into a COP test requires the whole contact and preparation history. The descending masses continue losing gravitational potential energy; compliant legs or a bending board can store and return work; active forces can contribute during the same interval. Chosen endpoint speeds leave these histories to be identified.

What made the question open. The support adds another intriguing feature. An ideal fixed pivot can deliver an impulse while doing zero work, because its point of application stays still. Momentum and work therefore follow different accounts. The board's rotation, elastic recoil, catch engagement and separation time connect the incoming motion to the outgoing flight.

Models with the same prescribed takeoff speed can require different active work and contact-force histories. Takeoff time, board recoil and compression history distinguish mechanisms that a final flight height leaves together. The associated hardware question organizes the force, motion and calibration records needed to make that distinction.

The physical idea. Let falling masses replace active performers, then use geometry, compliance and arrival timing to organize delivery. A spring receives work during one part of the motion and returns it while the receiver moves differently. A second arrival meets a board already rotating or deforming. Changing that timing gives a concrete route to explore.

The passive apparatus has its own model, separate from the human teeterboard. Its attraction is a directly testable COP: useful lift compared with every input needed to prepare, operate and restore a compact mechanical system.

1.1 Experiment: test COP over a full lever cycle

Figure 2 turns the launch into a COP measurement. Start before lifting the drivers; finish restored to the same ready state, with useful output retained outside the machine.

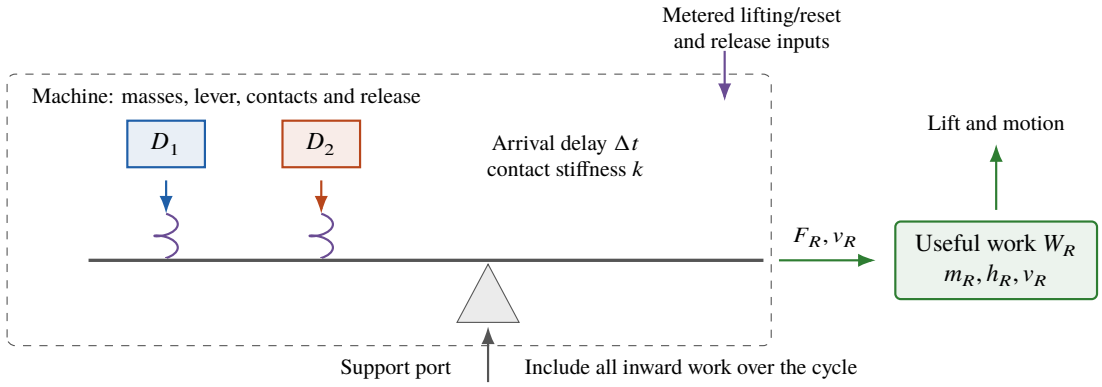


Figure 2. Lever COP experiment, schematic. Meter lifting/reset, release and support inputs; measure useful work at the outside receiver.

Compare complete cycles. Test rigid and compliant contacts, one and two drivers, and simultaneous and delayed arrivals Δt. Fix masses, heights and load across each configuration's repetitions; vary delay relative to contact duration. Include unloaded and loaded controls.

Measure the useful output. For a guided, rigid receiver of mass m_R, upward interface force $F_R = m_R\dot{v}_R + m_R g$ gives, without friction,

$$W_R = \int_{t_0}^{t_1} F_R v_R \, dt = m_R g \, \Delta h_R + \frac{m_R}{2}\Delta(v_R^2). \tag{2}$$

Synchronize force and motion at that interface. Receiver deformation and friction need their own terms. Isolate the delivered load and record return flow.

Meter the denominator. Each D_k is gross inward port work over the same full interval, defined on the final page:

$$\mathrm{COP}_L = \frac{W_R}{E_{\mathrm{in},L}}, \qquad E_{\mathrm{in},L} = D_{\mathrm{lift}} + D_{\mathrm{release}} + D_{\mathrm{support}} + D_{\mathrm{other}}. \tag{3}$$

D_{lift} includes lifting and restoration work; D_{release} supplies the release mechanism. Include moving-support work and every other inward transfer. A fixed ideal pivot has zero work. Check driver gravitational storage independently of input.

Restore and check the state. For a rigid lever centered at its fixed pivot, translating drivers and linear springs,

$$U_{\mathrm{mech}} = \sum_i \left(\frac{1}{2}m_i v_i^2 + m_i g h_i\right) + \frac{1}{2}I\dot{\theta}^2 + \sum_j \frac{1}{2}k_j\delta_j^2. \tag{4}$$

Here I is pivot inertia and δ_j spring deformation; the sums exclude the receiver. Check driver heights and velocities, lever angle, springs and latches, with flexural and thermal changes separately identified. Moving supports require their corresponding kinematics and stores.

Report W_R, $E_{\mathrm{in},L}$ and the COP lower bound. Pursue a bound above one with restored states and independent repetitions; contact histories connect timing to COP.

2 The switched transformer

A transformer with a tap offers several places to connect a receiver. The motivating analogy is a planet gear: rotation relative to its carrier differs from rotation relative to the ring. In the electrical counterpart, the chosen terminal pair determines the observed voltage. The return connection becomes a design variable for the COP question: can a timed sequence deliver more receiver work than all source, switching and reset inputs?

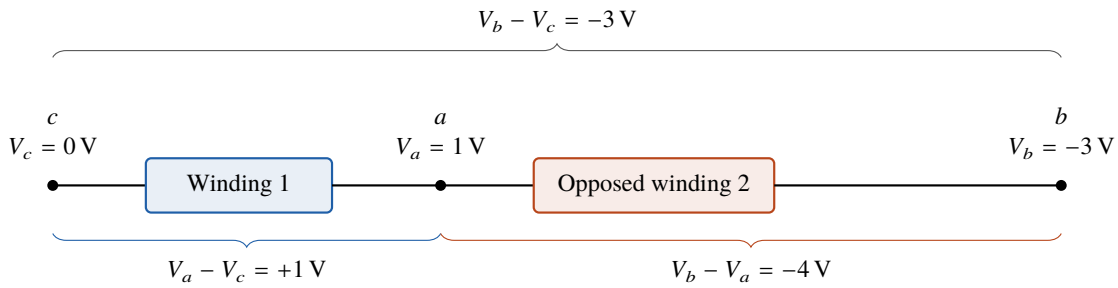


Figure 3. Same instantaneous ideal state, shown as a topology schematic. Opposed winding voltages add algebraically, giving two different receiver-terminal voltages.

The observation. Winding 1 joins terminals a, c; winding 2 joins b, a. On one electrical trajectory,

$$V_b - V_c = (V_b - V_a) + (V_a - V_c). \tag{5}$$

An ideal opposed winding ratio $n = -4$ means $V_b - V_a = n(V_a - V_c)$. With $V_a - V_c = 1\,\mathrm{V}$, Figure 3 therefore gives $-4\,\mathrm{V}$ across $b - a$ and $-3\,\mathrm{V}$ across $b - c$. The tempting idea is to use the two connections at different stages of a cycle.

A lightly loading meter reveals those voltages on essentially the same state. A receiver drawing current also changes the return path and subsequent motion. Each loaded connection has its own current history. Shifting every potential by the same numerical offset leaves the differences unchanged; moving the actual return wire creates the physical intervention.

What made the question open. Two fixed-connection descriptions leave the transition between them to be supplied. A winding can still carry current when its switch opens, and a capacitor can still hold voltage when a new connection is made. The current then follows a clamp, winding capacitance, an arc or another conducting path. Charge redistributes, and the selector's driver participates in the event.

The original descriptions retained current paths through their declared changes. Actual opening and tap commutation therefore required a fuller physical account. Later finite models represent particular transitions; the real switch and gap laws remain to be identified. One bounded complete-history comparison needed more supply in its switched case at matched receiver delivery, providing further designs with a concrete benchmark.

The physical idea. Connect the receiver where the evolving winding and capacitor states favor delivery, then reconnect for a different part of the cycle. Pursue a sequence whose useful output exceeds its complete measured input. The voltage comparison identifies a switching opportunity; integrating receiver and supply powers through the actual cycle determines its COP.

2.1 Experiment: test COP with physical tap commutation

Figure 4 tests whether moving the receiver return between a and c can produce COP above one. Use an isolated, current-limited low-voltage source, finite selector and clamp, and independently metered drivers.

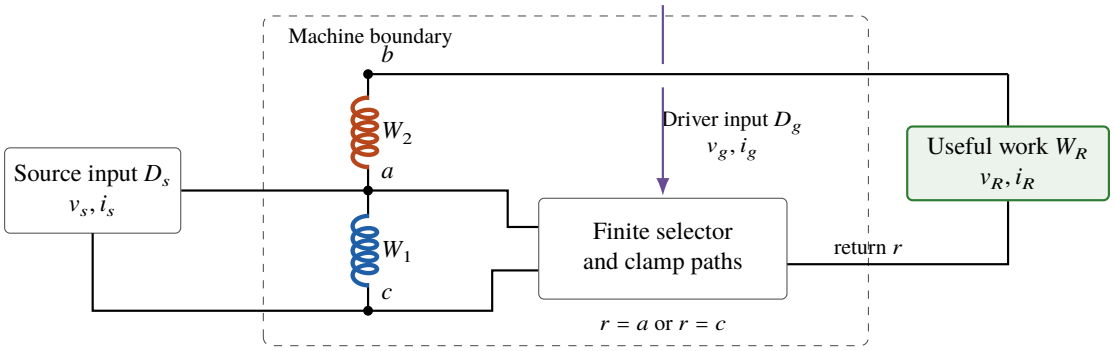


Figure 4. Transformer COP experiment, functional schematic. Source and driver supplies are inputs; the outside receiver supplies the useful-output measurement. Meter the entire cycle.

Choose a useful service. Compare fixed return at a, fixed return at c, and commutation at declared phases. Adjust a declared drive setting to match receiver work and delivery time at a fixed load, then compare complete input and COP. Include zero-current and finite-current switching.

Integrate output and every supply. If j_R enters the receiver at b,

$$W_R = \int_{t_0}^{t_1} (V_b - V_r) j_R \, dt, \qquad \mathrm{COP}_T = \frac{W_R}{D_s + D_g + D_{\mathrm{other}}}. \tag{6}$$

Here $D_s = \int_{t_0}^{t_1} [v_s i_s]_+ \, dt$ is the external source's gross input, D_g sums the independent driver supplies in the same way, and $[x]_+ = \max(x, 0)$. D_{other} includes every remaining inward transfer. Keep source return separate and include every supply's preparation and reset work.

Check the output independently. A characterized constant-capacitance receiver gives the ideal endpoint check

$$W_R = \frac{C_R}{2} \left(V_{R,f}^2 - V_{R,i}^2 \right). \tag{7}$$

Account separately for receiver leakage and dissipation where present. A resistor with calibrated calorimetry offers another check. Use synchronized voltage/current channels and a known dummy-load switching transient to check timing and bandwidth.

Finish the cycle. Start before energization from a declared ready state. Record switching, clamp conduction, relaxation and actual reset, ending at that same machine state. Check winding currents, capacitor voltages, controller state and magnetic/thermal history independently. The receiver remains outside; record any energy it sends back. Internal clamp transfer explains delivery and heating without adding a second boundary output.

Report W_R, total gross input and the COP lower bound for every configuration. A switched design can improve on a fixed tap while its absolute COP remains below one. The target is a lower bound above one with repeatable restored states; use the switching records to develop and independently retest that candidate.

3 The contact chain

A row of unequal masses and compliant contacts can pass a disturbance from one end to the other. Some masses reverse while others advance, and a light terminal mass can become fast. The COP ambition is to organize these local exchanges into useful receiver work exceeding the complete preparation and reset input.

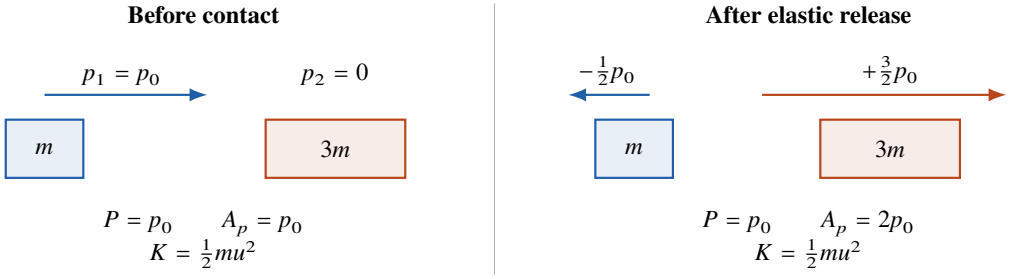


Figure 5. Exact two-mass elastic-contact illustration, with $p_0 = mu > 0$. Arrow lengths represent momenta; opposite motion doubles their magnitude sum while the signed total stays fixed.

The observation. An exact building block makes the effect visible. A mass m arrives at speed $u > 0$; a mass $3m$ is initially still. During a lossless Hooke contact of stiffness k, let δ be compression. The force laws $m\dot{v}_1 = -k\delta$, $3m\dot{v}_2 = k\delta$ and $\dot{\delta} = v_1 - v_2$ give, from $\delta(0) = 0$,

$$v_1 = \frac{u}{4}(1 + 3\cos\theta), \qquad v_2 = \frac{u}{4}(1 - \cos\theta), \qquad \theta = t\sqrt{\frac{4k}{3m}}. \tag{8}$$

At $\theta = \pi$, compression returns to zero and the masses separate. Their momenta are $-p_0/2$ and $3p_0/2$, as in Figure 5. The signed sum remains p_0, while the sum of magnitudes rises from p_0 to $2p_0$. Endpoint kinetic energy is $\frac{1}{2}mu^2$ in both cases.

For a longer free chain, the same distinction is captured by $A_p = \sum_j |p_j| = P + 2\sum_{p_j<0} |p_j|$, where $P = \sum_j p_j$. Equal-and-opposite internal forces preserve P; opposing motion can increase A_p. Kinetic energy, spring storage and receiver work provide complementary views of that motion.

What made the question open. A permanent spring can push and pull; a collision contact releases. The interacting pairs therefore change during motion. The starting descriptions used free compliant chains with zero initial gaps. Gaps introduce travel and change which events overlap. Branches divide routes, while restitution, plasticity and fixture motion change the contact physics. Increasing stiffness raises a further question about the limiting impulse and work.

The physical idea. Choose mass order and spacing so a terminal mass meets the next transfer at a useful moment. Loading that mass feeds a reaction back through the contacts. The two-mass illustration identifies one mechanism; the COP test extends it to a loaded chain and asks whether delivered work exceeds all launch, actuator and restoration inputs. Track every mass while pursuing that complete-cycle result.

3.1 Experiment: measure the loaded chain's COP

Use three unequal carts and spring bumpers, as in Figure 6. Compare launch and restoration input with useful receiver work. Mass order, gaps and branching are controls for pursuing COP above one.

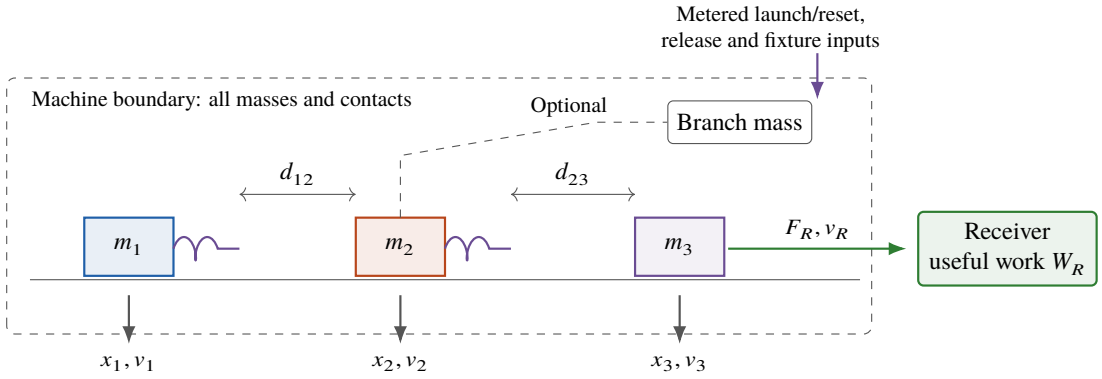


Figure 6. Chain COP experiment, schematic. Meter launch/reset, release and fixture inputs. The terminal contact delivers useful work to the outside receiver.

Compare loaded cycles. Test zero gaps against measured nonzero gaps d_{12}, d_{23}, reverse the mass order, and compare the line with a separately declared branch. Hold the prepared incidence fixed within each comparison; measure its preparation work. Compare unloaded, nominal and heavier receivers. A direct launch-to-receiver control can show what adding the chain changes.

Measure output and total input. With force and velocity at the same receiver interface, the primary result is

$$W_R = \int_{t_0}^{t_1} F_R v_R \, dt, \qquad \mathrm{COP}_C = \frac{W_R}{D_{\mathrm{launch}} + D_{\mathrm{release}} + D_{\mathrm{fixture}} + D_{\mathrm{other}}}. \tag{9}$$

Each D_k is gross inward port work over the complete cycle. Launch preparation and repositioning belong in D_{launch}; meter release-actuator and moving-fixture work separately. Include all other inward transfers. Check receiver work independently from its calibrated load and endpoint states.

Follow every internal store. In the unilateral Hooke model, $f_e = k_e[\delta_e]_+$, where $[x]_+ = \max(x, 0)$ and negative compression is an open gap. Integrating this contact force gives, for horizontal translation,

$$U_{\mathrm{mech}} = \sum_j \frac{p_j^2}{2m_j} + \sum_e \frac{1}{2} k_e [\delta_e]_+^2. \tag{10}$$

Gravity, rotation and fixture elasticity add stores when present. Damped or plastic contacts need identified force and release laws, including remaining deformation. Motion and force records reveal which contacts feed the receiver; exchanges inside the chain stay internal.

Close and repeat the history. Start before launch preparation with carts at rest in a declared configuration. Record delivery, separation, relaxation and actual repositioning to that state. Check every mass, spring, fixture, actuator and relevant thermal state. The receiver retains the delivered work outside the machine. Meter any return flow during restoration.

Report useful work, all gross input and the COP lower bound for each configuration. Pursue a bound above one with independent preparations and restored-state checks. Begin with resolvable soft contacts, then extend the COP test toward harder impacts.

4 The coupled-coil release

Two inductors exchanging through a capacitor provide an electrical counterpart to the contact chain. Flux linkages can acquire opposite signs while their signed sum stays fixed. Their magnitude sum then grows, suggesting a provocative next step: change the connection at a chosen moment and direct the evolving state into a receiver. The COP target is output exceeding all energization, switching and restoration input.

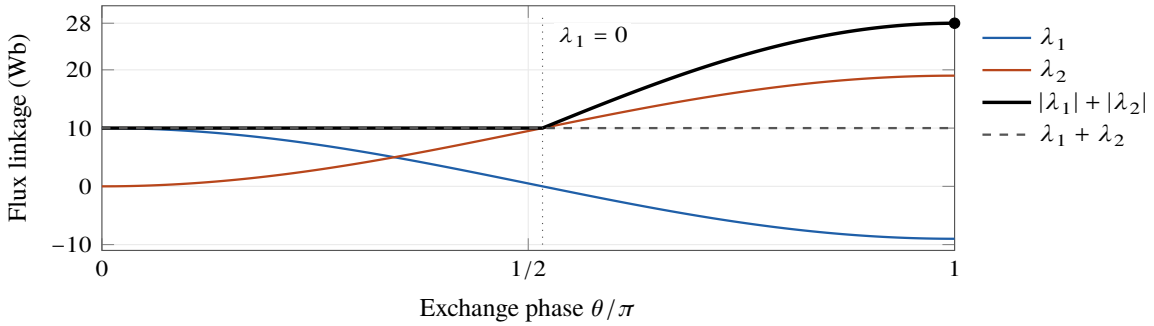


Figure 7. Exact connected-model curves for the stated zero-coupling illustration. Opposite linkage signs separate the growing magnitude sum from the fixed signed sum.

The observation. Take the zero-mutual-coupling limit, $L_1 = 1\,\mathrm{H}$, $L_2 = 19\,\mathrm{H}$ and $C > 0$, initially at $i_1 = 10\,\mathrm{A}$, $i_2 = 0$ and zero capacitor voltage. In the equal-and-opposite winding-voltage connection, the exact LC solution is

$$i_1 = \frac{1 + 19\cos\theta}{2}\,\mathrm{A}, \qquad i_2 = \frac{1 - \cos\theta}{2}\,\mathrm{A}, \qquad \theta = \omega t, \quad \omega^2 = \frac{1/L_1 + 1/L_2}{C}. \tag{11}$$

Figure 7 plots $\lambda_j = L_j i_j$. The first linkage crosses zero at $\theta = \arccos(-1/19)$. By $\theta = \pi$, the pair has moved from $(10, 0)$ to $(-9, 19)$ Wb: the signed sum stays 10 Wb while the magnitude sum reaches 28 Wb. The capacitor voltage is again zero at this endpoint. Independently evaluated magnetic storage is 50 J initially and $(81 + 19)/2 = 50\,\mathrm{J}$ finally. A growing linkage magnitude and redistributed stored energy occur together, giving a precise state from which to ask what a changed connection would deliver.

What made the question open. Opening a branch changes the equations that linked its winding voltages. The connected relation must be reconsidered on the new circuit. Ideal zero-current opening is a clean special case; finite-current opening introduces the switch gap, clamp, winding capacitance and possible arc. The physical current path, clamp release and complete circuit during transition become essential parts of the question.

The physical idea. Choose release time, coupling and winding orientation to steer transfer into a receiver. The connected example makes the motivating observation explicit. Actual release adds the COP experiment: choose a physical switching path, measure the useful delivery, and compare it with every input needed to create and repeat that release.

4.1 Experiment: test COP from energization to reset

Figure 8 tests preparation, exchange, release through S_1, S_R, relaxation and reset. A finite clamp carries the first coil's release current. The target is useful receiver work exceeding all supplied energy.

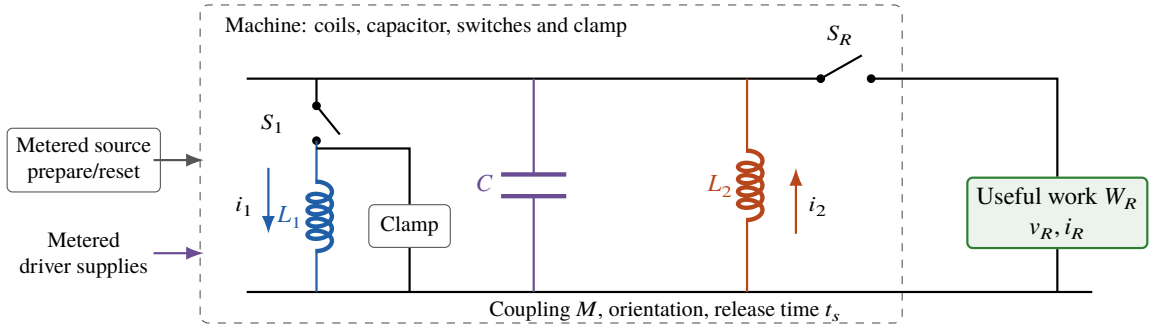


Figure 8. Coil-release COP experiment, functional schematic. Meter preparation/reset and driver supplies; integrate useful receiver work through delivery. Include all other inward ports.

Compare release strategies. At each fixed coupling and winding orientation, compare zero-current and declared finite-current release times t_s from the same prepared state. Test three coupling settings with geometry fixed within each cycle. Compare clamp and stray capacitance separately; include a zero-coupling release reference.

Make COP the measured result. For receiver voltage v_R and current i_R entering that receiver,

$$W_R = \int_{t_0}^{t_1} v_R i_R \, dt, \qquad \mathrm{COP}_{\mathrm{coil}} = \frac{W_R}{D_{\mathrm{source}} + D_{\mathrm{driver}} + D_{\mathrm{other}}}. \tag{12}$$

Each D_k is gross inward port work over the full history. Measure the actual source work of preparing the currents and any capacitor voltage, all independent driver supplies, and restoration work. Include other inward transfers in D_{other} and report source return separately. The initial magnetic store is an independent state check.

Account for both coils together. For fixed reciprocal linear inductances, consistent current directions and signed mutual inductance M,

$$U_M = \frac{1}{2}L_1 i_1^2 + M i_1 i_2 + \frac{1}{2}L_2 i_2^2, \qquad U_C = \frac{1}{2}C V_C^2. \tag{13}$$

Here $L_1, L_2 > 0$ and $M^2 < L_1 L_2$. With $\lambda_1 = L_1 i_1 + M i_2$ and $\lambda_2 = M i_1 + L_2 i_2$, differentiation gives $\dot{U}_M = i_1 \dot{\lambda}_1 + i_2 \dot{\lambda}_2$. Identify nonlinear magnetic and thermal states separately where the apparatus requires them.

Check useful delivery and restoration. A constant-capacitance receiver has ideal gain $C_R(V_{R,f}^2 - V_{R,i}^2)/2$; account for leakage or loss separately. Calibrated resistor calorimetry is another independent check. Resolve the switch and clamp transfers, then continue until the machine returns to its declared preparation-start state. Check currents, voltages, controller and magnetic/thermal states on both sides of release and after reset. Retain useful output outside.

Report W_R, all gross input and the COP lower bound for each strategy. Pursue a bound above one on independently prepared, restored cycles. Release and clamp traces explain the resulting COP.

The same test on every bench

Draw the machine boundary before measuring. At every crossing port k, take power P_k positive inward: vi electrically, $\mathbf{F} \cdot \mathbf{v}$ mechanically, or $\tau \cdot \omega$ rotationally. Effort and motion refer to the same station and frame. Input groups in each experiment are disjoint: count shared supplies once. Include signed heat and environmental transfers when present. For the complete interval $[a, b]$, define

$$W_k = \int_a^b P_k \, dt = D_k - X_k, \qquad D_k = \int_a^b \max(P_k, 0) \, dt, \qquad X_k = -\int_a^b \min(P_k, 0) \, dt, \tag{14}$$

$$E_{\text{in}} = \sum_k D_k, \qquad E_{\text{out}} = \sum_k X_k, \qquad r_E = \Delta U - E_{\text{in}} + E_{\text{out}}. \tag{15}$$

Here $\Delta U = U(b) - U(a)$ is evaluated independently from endpoint states; r_E is the signed balance residual. Positive r_E denotes outward excess relative to measured inputs and store change. Heat retained inside is a store change; heat crossing the boundary is a transfer. Count each once.

Let $E_{\text{useful}} = W_R$ be net useful work delivered to the declared outside receiver, after any return flow. All inward receiver transfers also enter E_{in}. Report source return and unwanted heat separately. With a resolved, positive input denominator, the ratio and its conservative measurement bound are

$$\text{COP} = \frac{E_{\text{useful}}}{E_{\text{in}}}, \qquad \text{COP}_{\text{lower}} = \frac{E_{\text{useful}} - B_{\text{out}}}{E_{\text{in}} + B_{\text{in}}}. \tag{16}$$

The nonnegative bounds B_{in} and B_{out} must have independently established simultaneous coverage. Use this lower bound when $E_{\text{useful}} - B_{\text{out}} > 0$. Declare coverage and an input floor before evaluation; an unresolved or smaller input makes the ratio undefined. Identifying the ports and stores gives this bound its physical meaning.

Measure the whole history. Start before preparation in a reproducible ready state; finish restored to that state with useful output retained outside. Timestamp preparation, operation, switching/release, relaxation and actual reset without gaps or double counting. Split integration at events and retain both event-side states. Measure finite transitions or separately justify any impulse work. Verify mechanical, electrical, magnetic, controller and thermal endpoint states as applicable. Measure reset work directly and bound any remaining depletion independently. Use the same boundary for every comparison and every repetition.

Calibrate and compare. Establish gains, offsets, timing skew, bandwidth and correlated errors with known loads or motions. Use simultaneous channels, independent receiver checks and event-resolved integration checks. Finer acquisition must resolve the actual events. Predeclare drive levels, receiver loads, independent preparations and controls; retain failed starts and both residual signs. Assess measurement, integration and model uncertainty separately.

Follow the interesting result. A lower COP bound above one, supported by the boundary and endpoint checks, earns independent repetition. An unexplained deficit also points to physics worth investigating. Keep exploratory adjustments separate from fresh confirmation measurements. Start with the lever and transformer, then carry the instruments into the contact chain and coil release. Build the strongest experiment, follow the unexpected result, and make every apparent win survive another meter and another independent preparation.